Harmonic Oscillator in a Lennard-Jones Fluid: integration speed of velocity Verlet and r-NAPA

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Abstract

This work investigates the relative efficiency of the velocity Verlet (VV) and reversible numerical analytical propagator (r-NAPA) algorithms in simulating a 3D system of N spherical particles (mass m, diameter d) confined inside a cubic volume V under periodic boundary conditions. Interparticle dynamics are governed by pairwise Lennard-Jones interactions, with one particle further subjected to a harmonic potential of elastic constant k. C++ implementations reveal that the speedup of r-NAPA over VV scales as $\sqrt{k}$.

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1 Introduction

1.1 Harmonic oscillator in a Lennard-Jones fluid

The system of interest consists of N spherical particles of mass m and diameter d confined inside a cubic volume V under periodic boundary conditions. Interparticle dynamics are governed by pairwise shifted Lennard-Jones interactions with a cut-off, with one particle further subjected to a harmonic potential, see Fig.1.

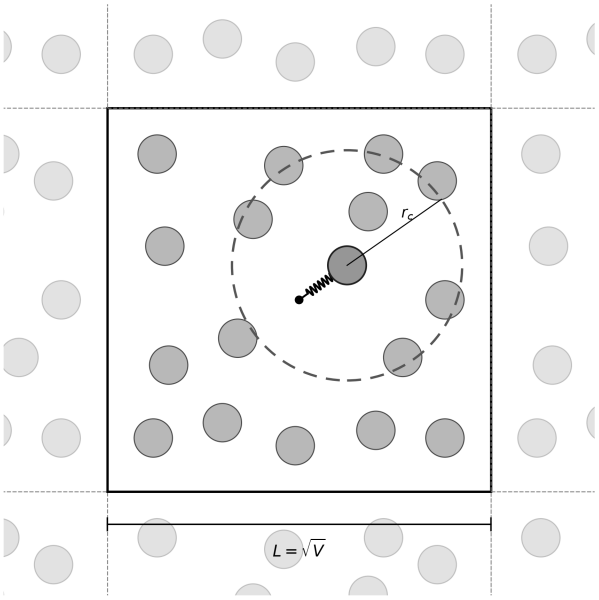


Figure 1: Schematic representation of the system: N particles of diameter d in a box of volume V with periodic boundary conditions. Lennard-Jones interaction with a cut-off radius r_c and one particle subjected to a harmonic potential with elastic constant k.

The Hamiltonian of this system can be written as follows:

$$H(\mathbf{r}_{1,\dots N}, \mathbf{v}_{1,\dots,N}) = \sum_{i=1}^{N} \frac{1}{2} m \mathbf{v}_i^2 + \frac{1}{2} \sum_{i,j=1}^{N} U_{\mathrm{LJ}}(r_{ij}) + \frac{1}{2} k \mathbf{r}_1^2 \tag{1}$$

where $\mathbf{r_i}$ is a particle's position vector and $\mathbf{v_i}$ its velocity, $r_{ij} = |\mathbf{r_i} - \mathbf{r_j}|$, σ determines the minimum distance at which the Lennard-Jones potential is zero, ϵ determines the depth of the potential well and

$$U_{\mathrm{LJ}}(r_{ij}) = \begin{cases} 4\varepsilon \left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6} \right] - 4\varepsilon \left[\left(\frac{\sigma}{r_c}\right)^{12} - \left(\frac{\sigma}{r_c}\right)^{6} \right], & r_{ij} \leq r_c \\ 0, & r_{ij} > r_c \end{cases} \tag{2}$$

1.2 Integration algorithms

Velocity Verlet Velocity Verlet is a second-order, symplectic, time-reversible integration algorithm that updates each particle's position and velocity according to the following steps:

1. **Position update:** compute the new positions $r_i(t + \Delta t)$ using starting positions, velocities and accelerations:

$$\mathbf{r_i}(t + \Delta t) = \mathbf{r_i}(t) + \mathbf{v_i}(t)\Delta t + \frac{1}{2}\mathbf{a_i}(t)\Delta t^2 \tag{3}$$

$$\tag{4}$$

where $\mathbf{a_i}(t)$ for the system of interest will be

$$\mathbf{a_i}(t) = -\frac{1}{m}\sum_{j\neq i}\frac{dU_{LJ}(r_{ij})}{dr_{ij}}\hat{\mathbf{r}}_{ij} - \delta_{1i}\frac{k\mathbf{r}_i}{m} \tag{5}$$

where δ_{1i} is a Kronecker delta that is 1 when $i = 1$ and 0 for all other particles, and $\hat{\mathbf{r}}_{ij} = \frac{\mathbf{r_i}-\mathbf{r_j}}{|\mathbf{r_i}-\mathbf{r_j}|}$.

2. **Force evaluation:** calculate the new accelerations $a_i(t + \Delta t)$ using the updated positions $r_i(t + \Delta t)$

3. **Velocity update:** compute the new velocities $v_i = t + \Delta t$:

$$v_i(t + \Delta t) = v_i(t) + \frac{1}{2}[a_i(t) + a_i(t + \Delta t)]\Delta t \tag{6}$$

Performing a numerical integration of a system of N interacting particles with this algorithm thus requires $\mathcal{O}(N^2)$ operations for each time increment, dictated by the $\frac{N(N-1)}{2}$ pairwise interactions.

r-NAPA The Reversible numerical analytical propagator algorithm (r-NAPA) is a second-order, symplectic, time-reversible integration algorithm specialized for systems characterized by interactions of different time scales through the introduction of multiple integration time steps. In this case it separates the time evolution due to the harmonic interaction, of time scale $\tau_k \propto \sqrt{m/k}$, and the one due to the Lennard-Jones interaction $\tau_{LJ} \propto \sigma\sqrt{m/\epsilon}$.

Defining Δt_k as the time increment relative to the harmonic potential, and Δt_{LJ} the increment relative to the Lennard-Jones potential, the integration follows the steps below:

1. **Slow force half-step:** perform a VV integration for all particles over a half slow time step $\frac{\Delta t_{\mathrm{LJ}}}{2}$, driven exclusively by the slow Lennard-Jones accelerations $\mathbf{a}_{i,\mathrm{LJ}}$:

$$\mathbf{r}_i\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\frac{\Delta t_{\mathrm{LJ}}}{2} + \frac{1}{2}\mathbf{a}_{i,\mathrm{LJ}}(t)\left(\frac{\Delta t_{\mathrm{LJ}}}{2}\right)^2 \tag{7}$$

$$\mathbf{v}_i\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right) = \mathbf{v}_i(t) + \frac{1}{2}\left[\mathbf{a}_{i,\mathrm{LJ}}(t) + \mathbf{a}_{i,\mathrm{LJ}}\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right)\right]\frac{\Delta t_{\mathrm{LJ}}}{2} \tag{8}$$

2. **Fast force integration:** perform n VV steps of size Δt_k, such that $n\Delta t_k = \Delta t_{LJ}$ exclusively for the bound particle ($i = 1$), driven solely by the fast harmonic potential $\mathbf{a}_{\mathrm{k}}$. Iterate for $m = 0, \ldots, n-1$ starting from $\tau_0 = t + \frac{\Delta t_{\mathrm{LJ}}}{2}$ (where $\tau_{m+1} = \tau_m + \Delta t_k$):

$$\mathbf{r}_1(\tau_{m+1}) = \mathbf{r}_1(\tau_m) + \mathbf{v}_1(\tau_m)\Delta t_k + \frac{1}{2}\mathbf{a}_k(\tau_m)\Delta t_k^2 \tag{9}$$

$$\mathbf{v}_1(\tau_{m+1}) = \mathbf{v}_1(\tau_m) + \frac{1}{2}\left[\mathbf{a}_k(\tau_m) + \mathbf{a}_k(\tau_{m+1})\right]\Delta t_k \tag{10}$$

3. **Slow force half-step:** complete the integration by performing a second VV integration for all particles over another half slow time step $\frac{\Delta t_{\mathrm{LJ}}}{2}$ using the slow Lennard-Jones potential, updating the system state from $t + \frac{\Delta t_{\mathrm{LJ}}}{2}$ to $t + \Delta t_{\mathrm{LJ}}$:

$$\mathbf{r}_i(t + \Delta t_{\mathrm{LJ}}) = \mathbf{r}_i\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right) + \mathbf{v}_i\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right)\frac{\Delta t_{\mathrm{LJ}}}{2} + \frac{1}{2}\mathbf{a}_{i,\mathrm{LJ}}\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right)\left(\frac{\Delta t_{\mathrm{LJ}}}{2}\right)^2 \tag{11}$$

$$\mathbf{v}_i(t + \Delta t_{\mathrm{LJ}}) = \mathbf{v}_i\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right) + \frac{1}{2}\left[\mathbf{a}_{i,\mathrm{LJ}}\left(t + \frac{\Delta t_{\mathrm{LJ}}}{2}\right) + \mathbf{a}_{i,\mathrm{LJ}}(t + \Delta t_{\mathrm{LJ}})\right]\frac{\Delta t_{\mathrm{LJ}}}{2} \tag{12}$$

As well as for VV, performing a numerical integration of a system of N interacting particles with this algorithm requires $\mathcal{O}(N^2)$ operations for each time increment, dictated by the $\frac{N(N-1)}{2}$ pairwise interactions. However, the separation of timescales significantly alters the computational efficiency of the integration.

In standard velocity Verlet, numerical stability dictates that the integration time step Δt must be bounded by the fastest characteristic timescale in the system, namely the period of the harmonic oscillator τ_k (typically requiring $0.001 \leq \Delta t/\tau_k \leq 0.01$).

In contrast, r-NAPA decouples these degrees of freedom, allowing the slow Lennard-Jones fluid interactions to be integrated using a large time step Δt_{LJ} dictated by the fluid timescale τ_{LJ}, while applying the smaller time step Δt_k exclusively to the fast harmonic degree of freedom. Thus for an integration over a time interval T r-NAPA introduces a speedup with respect to VV that amounts to $S = \frac{\Delta t_{LJ}}{\Delta t_k} \propto \sigma\sqrt{\frac{k}{\epsilon}}$.